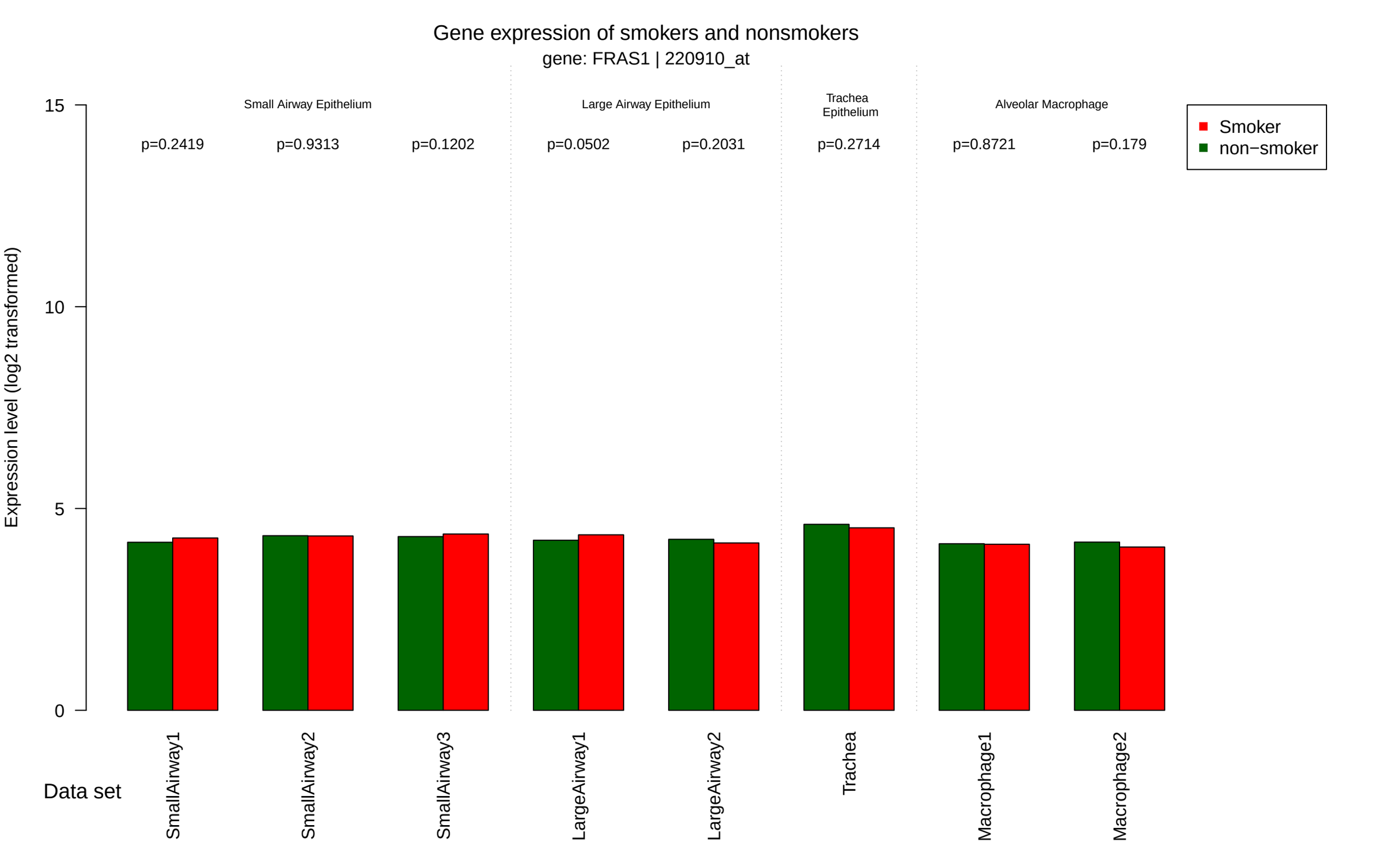
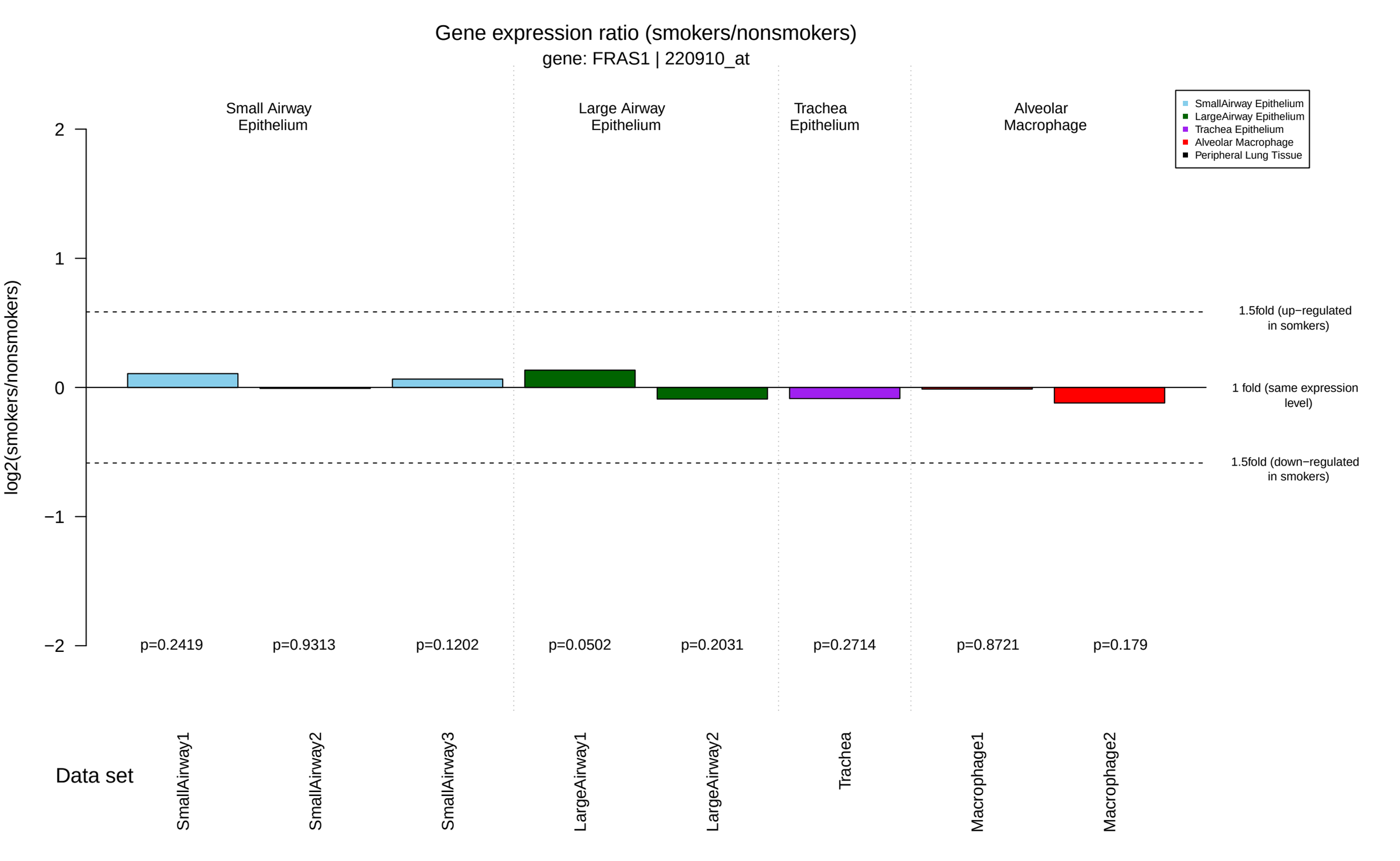
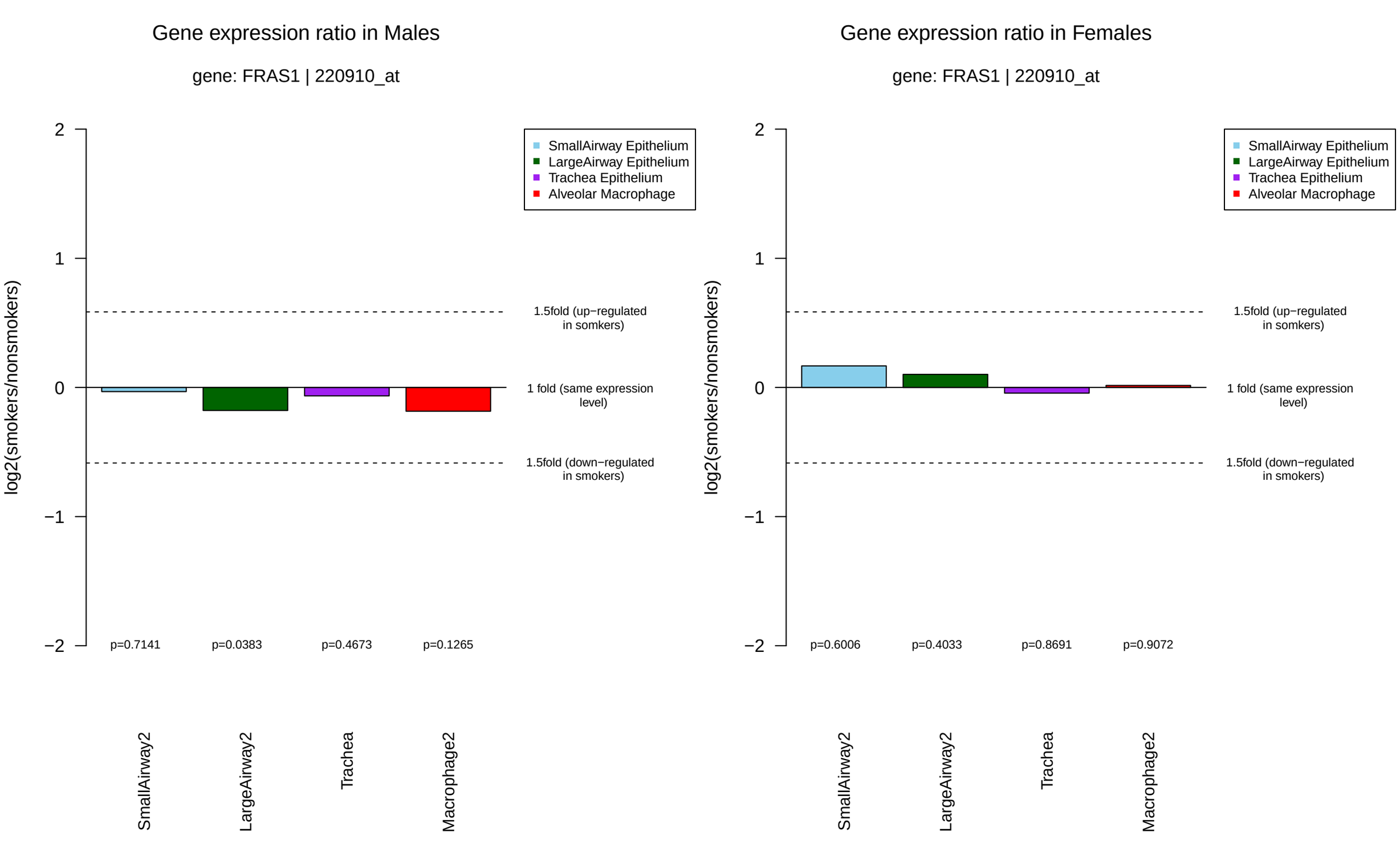
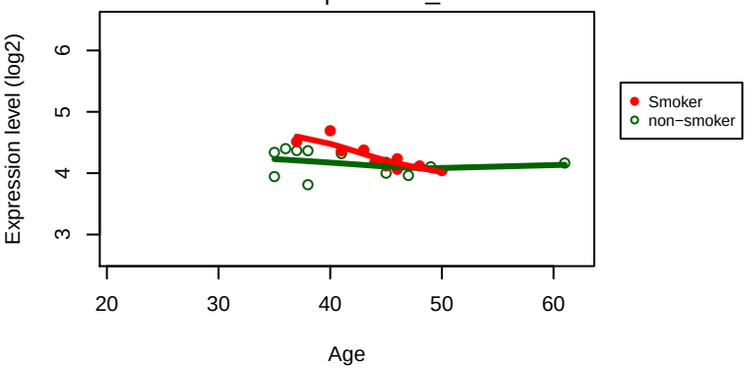
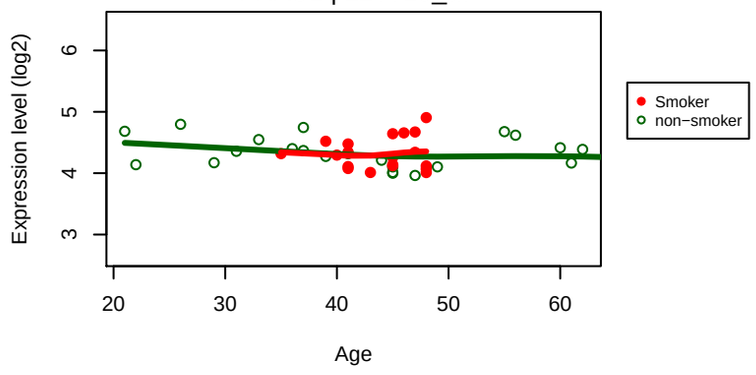
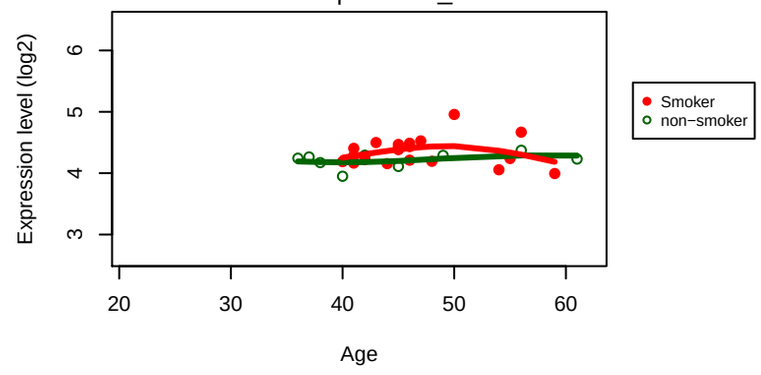

Gene expression of smokers and nonsmokers
gene: FRAS1 | 220910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2419
p=0.9313
p=0.1202
p=0.0502
p=0.2031
p=0.2714
p=0.8721
p=0.179
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

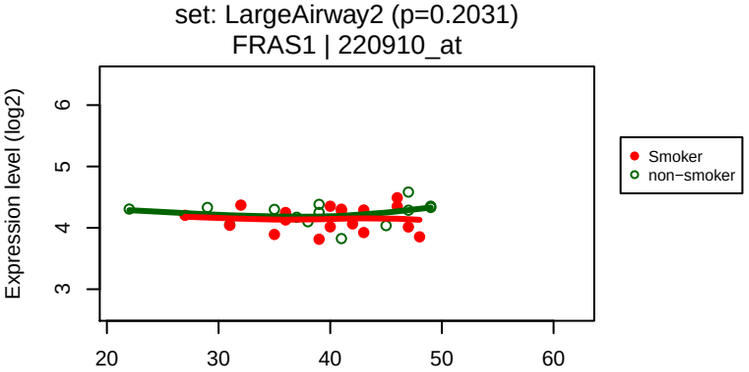

Gene expression ratio (smokers/nonsmokers)
gene: FRAS1 | 220910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2419
p=0.9313
p=0.1202
p=0.0502
p=0.2031
p=0.2714
p=0.8721
p=0.179
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

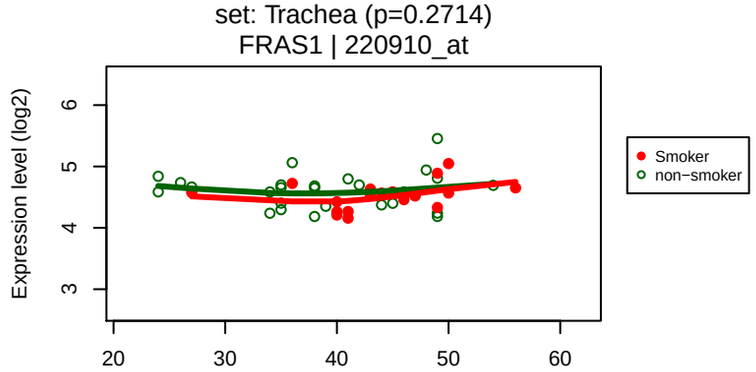

Gene expression ratio in Males
gene: FRAS1 | 220910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7141
p=0.0383
p=0.4673
p=0.1265
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRAS1 | 220910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6006
p=0.4033
p=0.8691
p=0.9072
SmallAirway2
LargeAirway2
Trachea
Macrophage2

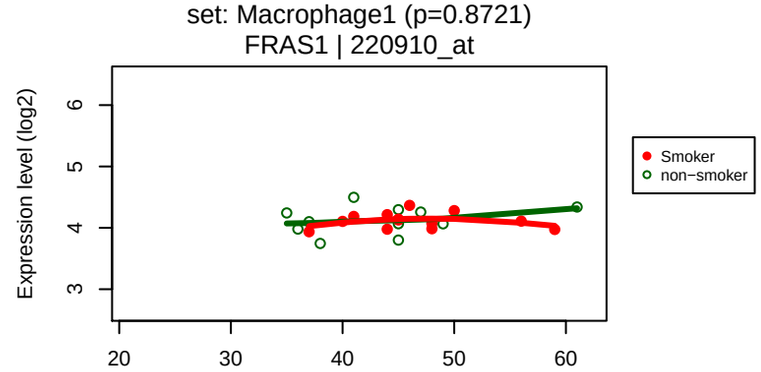

set: SmallAirway1 (p=0.2419)
FRAS1 | 220910_at
Expression level (log2)
Age
Smoker
non−smoker

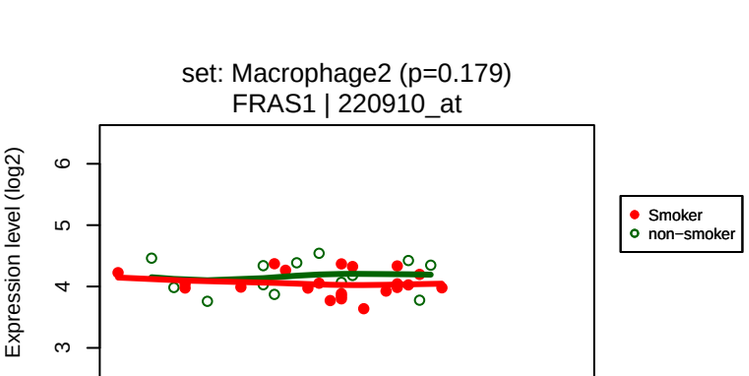

set: SmallAirway2 (p=0.9313)
FRAS1 | 220910_at
Expression level (log2)
Age
Smoker
non−smoker

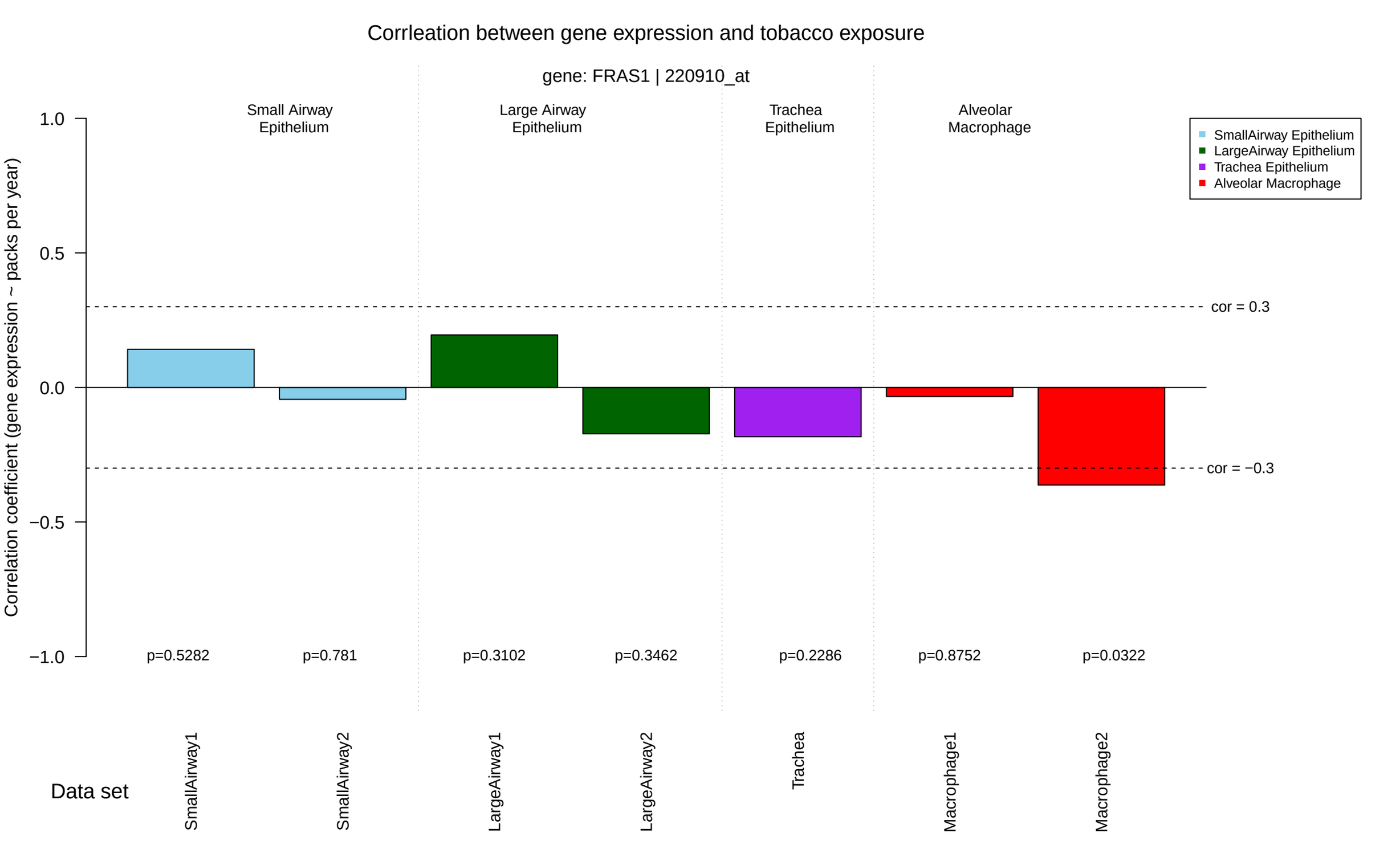

set: LargeAirway1 (p=0.0502)
FRAS1 | 220910_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2031)
FRAS1 | 220910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2714)
FRAS1 | 220910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8721)
FRAS1 | 220910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.179)
FRAS1 | 220910_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRAS1 | 220910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5282
p=0.781
p=0.3102
p=0.3462
p=0.2286
p=0.8752
p=0.0322
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2